



Introduction to IT 2025



Introduction to LaTeX
Beamer Presentations – Part 1

What is Beamer?

- **Beamer** is a LaTeX document class for creating presentation slides.

What is Beamer?

Key Features:

- Professional-looking presentations
- Built on LaTeX - all LaTeX commands available
- Automatic navigation and structure
- Consistent styling across slides
- Mathematical formulas support
- Wide variety of themes

What is Beamer?

So :) It is perfect for:

⇒ Academic presentations,

⇒ Technical talks,

⇒ Scientific conferences

Setting Up a Beamer Document

```
\documentclass{beamer}
```

```
\usetheme{Boadilla}
```

```
\begin{document}
```

```
% Your slides go here
```

```
\end{document}
```

Document Preamble

```
\documentclass{beamer}
```

```
\usetheme{Boadilla}
```

```
\title{My Presentation}
```

```
\subtitle{Using Beamer}
```

```
\author{Joe Bloggs}
```

```
\institute{University of ShareLaTeX}
```

```
\date{\today}
```

Document Preamble

```
\begin{document}
```

```
% Slides here
```

```
\end{document}
```

Creating the Title Page

```
\begin{document}
```

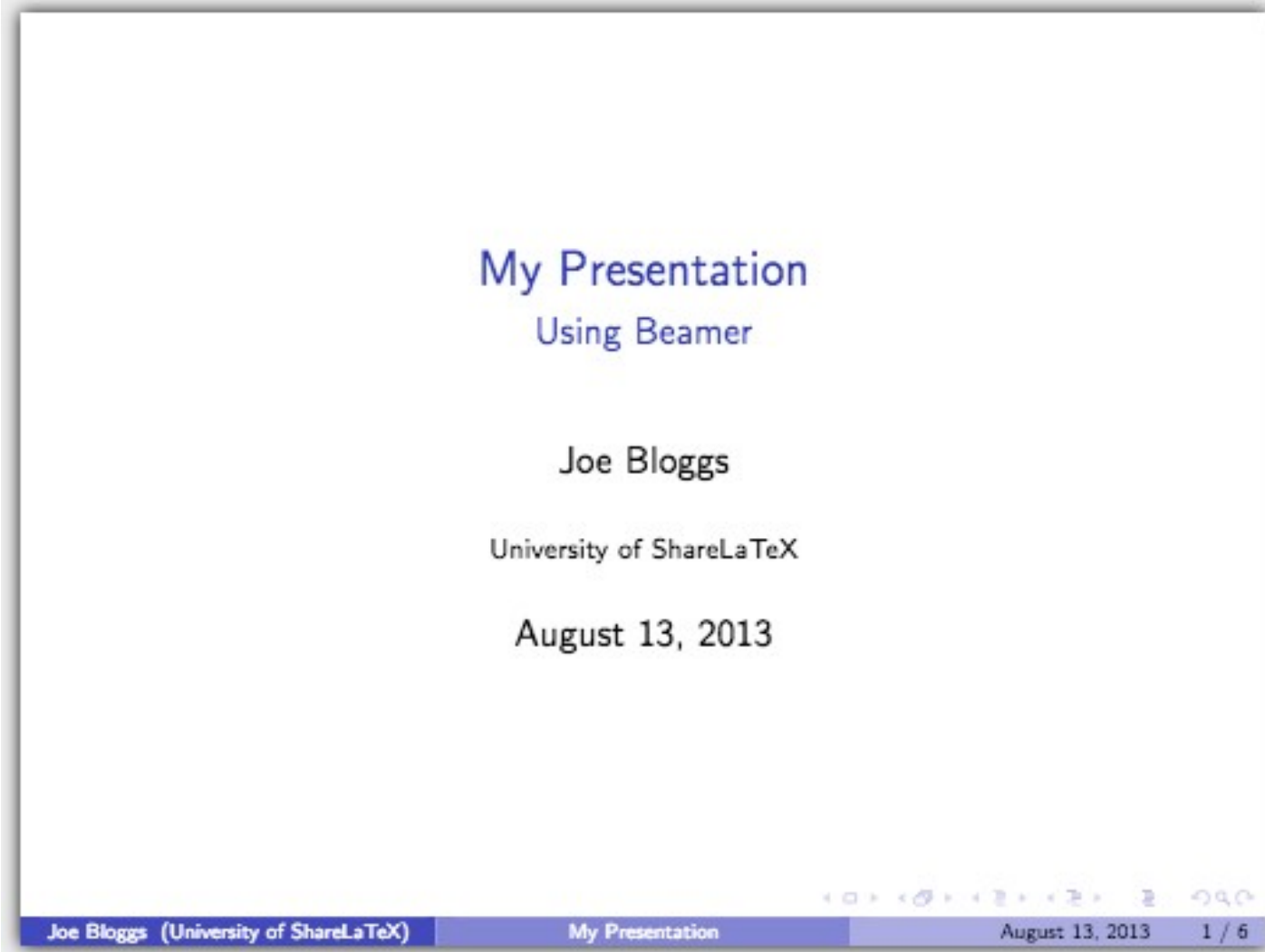
```
\begin{frame}
```

```
\titlepage
```

```
\end{frame}
```

```
\end{document}
```


Creating the Title Page



My Presentation
Using Beamer

Joe Bloggs

University of ShareLaTeX

August 13, 2013

Navigation icons: back, forward, search, etc.

Joe Bloggs (University of ShareLaTeX)	My Presentation	August 13, 2013	1 / 6
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Understanding Frames

- Frames are the basic building blocks of Beamer presentations

Understanding Frames

```
\begin{frame}
```

```
\frametitle{My Slide Title}
```

Content goes here...

```
\end{frame}
```

Structuring Your Presentation

```
\section{Introduction}
```

```
\subsection{Background}
```

```
\begin{frame}
```

```
\frametitle{Background Information}
```

Content here...

```
\end{frame}
```

Structuring Your Presentation

```
\subsection{Motivation}
```

```
\begin{frame}
```

```
\frametitle{Why This Topic?}
```

```
More content...
```

```
\end{frame}
```

Adding Content to Frames

```
\section{Section 1}
```

```
\subsection{sub a}
```

Adding Content to Frames

```
\begin{frame}
```

```
\frametitle{Title}
```

```
Lorem ipsum dolor sit amet,  
consectetur
```

```
adipiscing elit, sed do eiusmod  
tempor
```

```
incididunt ut labore et dolore magna  
aliqua
```

Adding Content to Frames

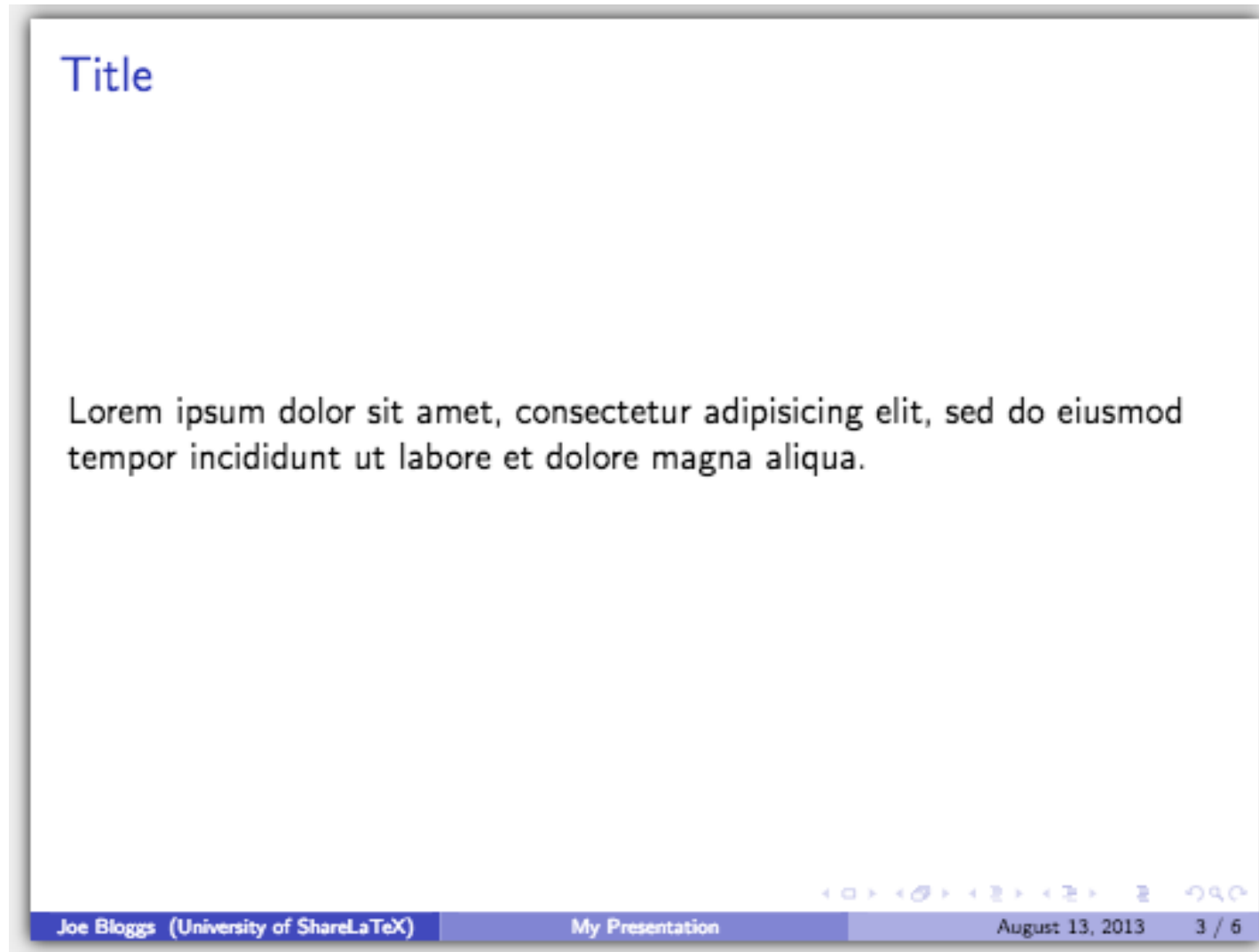


Table of Contents

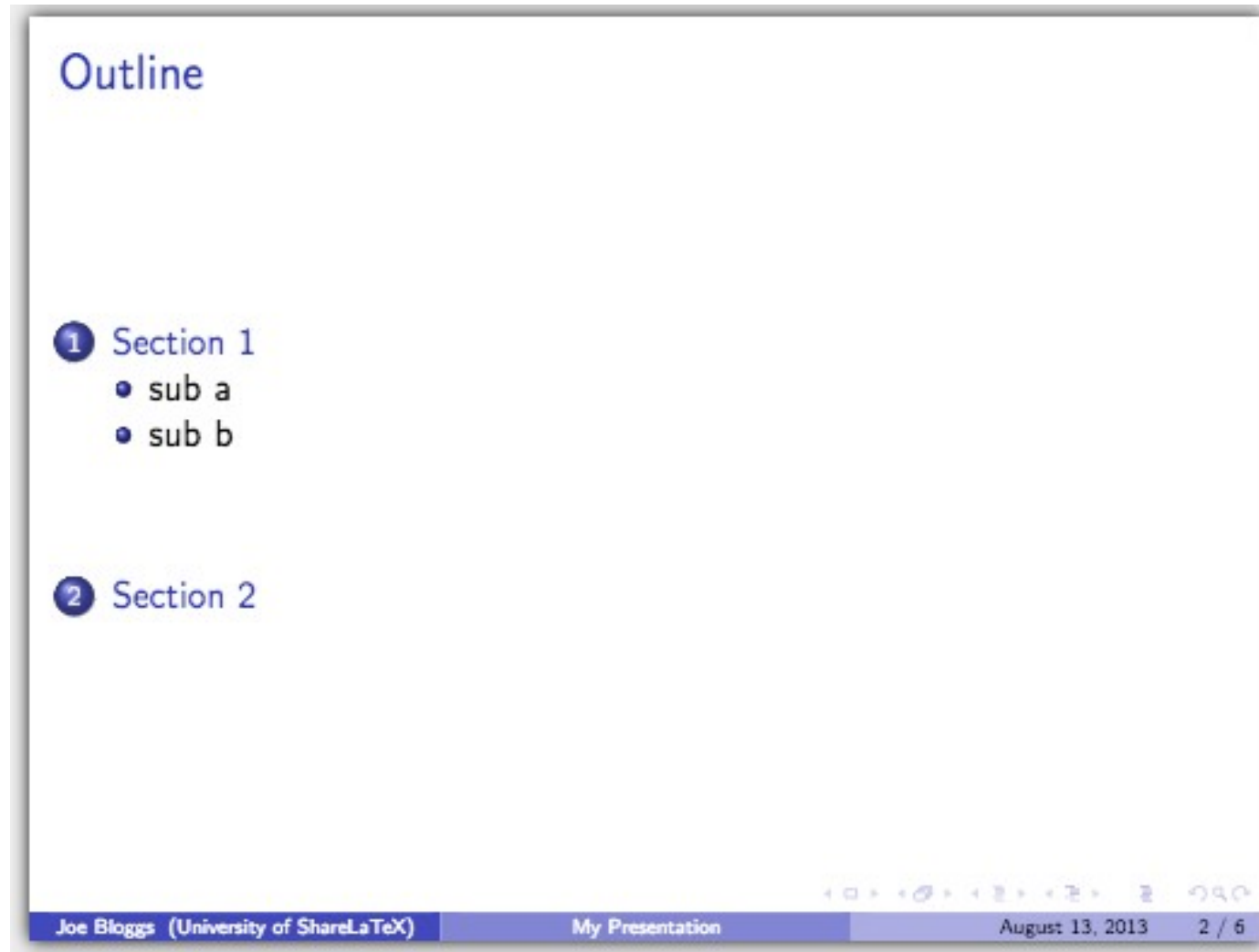
```
\begin{frame}
```

```
\frametitle{Outline}
```

```
\tableofcontents
```

```
\end{frame}
```

Table of Contents



Outline

- 1 Section 1
 - sub a
 - sub b
- 2 Section 2

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This is a screenshot of a presentation slide titled 'Table of Contents'. The slide features a white background with a blue header area. The title 'Table of Contents' is displayed in a large, bold, blue font at the top. Below the title, the word 'Outline' is written in a smaller, blue font. The outline consists of two main sections: '1 Section 1' and '2 Section 2'. Section 1 is further detailed with two sub-points: '• sub a' and '• sub b'. At the bottom of the slide, there is a blue footer bar containing the text 'Joe Bloggs (University of ShareLaTeX)', 'My Presentation', 'August 13, 2013', and '2 / 6'. Above the footer bar, there are several small, light blue navigation icons.

Complete Example

- Check the `sample_slide.tex`

Automatic Features

Beamer automatically adds:

- Footer Bar (on every slide):
 - Author name
 - Institute
 - Presentation title
 - Date
 - Frame/slide number

Automatic Features

Beamer automatically adds:

- Navigation Buttons (bottom right):
 - Navigate between slides
 - Jump between frames
 - Move through subsections/sections
 - Go to first/last slide
 - Search functionality

Navigation Buttons Explained

1. **Slide navigation**   - Move one slide at a time
2. **Frame navigation** - Jump between frames
3. **Subsection navigation** - Navigate subsections
4. **Section navigation** - Jump between sections
5. **Document navigation** - First/last slide

Viewing Your Presentation

Steps to present:

1. Compile your `.tex` file
2. Download/open the generated PDF
3. Put PDF reader in fullscreen mode
4. Use navigation buttons or keyboard

Tip: Most PDF readers support presentation mode with keyboard shortcuts!

Best Practices

DO:

- Use clear, descriptive frame titles
- Organize content with sections/subsections
- Keep text concise on each slide
- Use the table of contents for longer presentations

Best Practices

DONOT:

- Overcrowd slides with too much text
- Forget to set title/author information
- Neglect to structure with sections

Common Themes Preview

Some popular Beamer themes:

- **Boadilla** - Clean and professional
- **Madrid** - Simple with sidebar
- **AnnArbor** - Blue theme with header
- **CambridgeUS** - Red accents
- **Berlin** - Mini navigation
- **Copenhagen** - Blue gradient



QA

Essential Commands

```
\documentclass{beamer}% Set document  
class
```

```
\usetheme{ThemeName} % Choose theme
```

```
\title{Title} % Set title
```

```
\author{Name} % Set author
```

```
\date{\today} % Set date
```

Essential Commands

`\begin{frame} ... \end{frame}` % Create slide

`\frametitle{Title}` % Frame title

`\titlepage` % Title page

`\tableofcontents` % TOC

`\section{Name}` % Section

`\subsection{Name}` % Subsection